

ARNAV GOEL

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EDUCATION

National University of Singapore

Aug 2023 - May 2027 [Expected]

Bachelor of Computing in Computer Science, Honours (GPA: 4.14/5.00)

Final Year Project: Developing a watchOS and iOS application for palliative care, using AI to analyse wearable vitals and camera-based movement cues against patient-specific baselines to flag early deterioration.

Interesting Modules: Responsible AI: From Algorithms to Impact and Conversational User Interface

Teacher Assistant for Introduction to Software Engineering and Spatial Computing: Design and Development.

TECHNICAL SKILLS

Programming Languages: Java, Python (NumPy, Pandas, PyTorch), Swift, R, JavaScript, TypeScript, Kotlin.

Web Development and Database: React.js, Node.js, Next.js, HTML, CSS, Material UI, Figma, MySQL, Firebase.

AI/ML: RAG, Agentic Workflows, Spec Driven Development, Machine Learning, Natural Language, Computer Vision.

Tools & Infrastructure: Git, CI/CD Pipeline, Docker, Bash/Shell scripting, AWS, Cloudflare, Tableau, Dataiku.

WORK EXPERIENCE

Johnson & Johnson, Singapore

Jan 2026 – July 2026

Data & Digital Business Analyst Intern [2 Appreciation Awards]

- Spearheaded the development of a Graph Qdrant RAG system with observability, reducing hours spent searching and validating information across enterprise PDFs by surfacing evidence-backed answers with traceable citations.
- Directed the development of a BA-first AI workflow platform for internal product discovery, automating multi-role workflow generation and downstream document updates designed to reduce weeks of manual effort to hours.
- Evaluated and developed AI agent PoCs for integration into existing business workflows, while designing Dataiku automations for repetitive scripting tasks to improve operational efficiency and reduce operating costs.
- Drove rapid prototyping and PoC development for internal Shark Tank pitches and J&J Hackathon initiatives, collaborating with teams to turn early-stage ideas into scalable and maintainable AI-ready solutions.

Interactive 3D, Singapore

May 2025 – Jan 2026

Software Engineer Intern

- Led an agile team of three and coordinated with NUH to digitise existing stroke rehab assessments, building multiplatform applications for Apple Vision Pro and iPad in Swift/Swift UI, to drive real-time recovery analytics.
- Developed a patent-pending 3D hand-eye coordination assessment tool on Vision Pro for quantifying hand dexterity, incorporating patient testing feedback and clinical requirements to support integration into healthcare workflows.
- Analysed and aggregated 15,000 data points through Python queries using Pandas and Matplotlib to process datasets, visualise key trends, and deliver actionable insights for patient recovery optimization.

Institute for Applied Sciences and Educational Technology (NUS), Singapore

Nov 2023 – Jun 2024

Software Engineer

- Designed a full-stack web application using React.js, JavaScript, and MySQL, employing OOP principles to automate coursework workflows, reducing administrative workload by 30% and boosting student engagement.
- Developed MySQL schema models and indexes to meet real-world requirements and boost query performance, and implemented complex, multi-table SQL queries to generate detailed user summaries, saving instructor's time.
- Collaborated closely with the client to translate user needs into product requirements and deliver a reliable application through unit and integration testing, robust error handling, and security scanning.

LightBlue Environmental Consulting, Bangkok (NUS Global Internship Program)

May 2024 – Nov 2024

UX/UI Designer Intern at a Food Waste Consulting Startup

- Coordinated with international cross-functional teams to develop customer personas and design user-friendly web and mobile prototypes in Figma, ensuring alignment with customer needs and boosting user satisfaction.
- Conducted competitor analysis and business research to benchmark best practices, deepening my understanding of business operations, market dynamics, and revenue drivers, and guided company on design change decisions.
- Refined interpersonal and project-management skills by adapting to a fast-paced startup in a new country and collaborating with a multicultural team (such as European, American, Thai) to deliver projects on tight deadlines.

PROJECTS

3D Singapore Discovery Platform (amblefinds.com): Applied speckit to launch a public beta serving 100+ monthly users, engineering evidence-backed web-scraping pipelines for events, restaurants, and live promotions using OneMap and TinyFish, and integrating the OpenAI Realtime API for voice-based natural-language discovery.

Agent-First F&B Procurement: Built a stateful AI agent system, integrating OpenAI APIs with Gmail, PostgreSQL, and Telegram to plan and execute multi-step workflows across inventory requests, supplier negotiation, delivery verification, and invoice reconciliation, with tool calling, exception escalation, and human-in-the-loop approvals.

Privacy-First Live Streaming App: Built an Android app using Java and on-device computer vision and OCR to cover faces, QR codes, private words and custom privacy zones in real time, ensuring only protected video is encoded and sent to live-streaming platforms such as TikTok through RTMP.

On-Device AI Low-Vision Assistant: Working with the Singapore Association of the Visually Handicapped (SAVH), GDS, NUH and low vision patients to identify pain points and translate them into accessible iPhone solutions, including Singapore currency recognition and step-by-step household-appliance guidance using Swift UI, Core ML, and Vision.

TRIBE v2 Music Optimization: Built a Python and PyTorch pipeline that uses predicted cortical responses to evaluate 21 music-production variations across seven controls and optimize a hip-hop track for a custom sleep-alignment objective reducing tempo, loudness and increasing duration.

EXTRA-CURRICULAR AND ADDITIONAL INFORMATION

Leadership Experience

- Appointed as Prince George's Park Residence Cluster Leader, responsible for hosting events with 15 floor residents and overseeing administrative duties such as resident safety and fire drills.
- Co-organised the world's first university hackathon dedicated to Vision Pro development, managing website development and mentoring student participants throughout the event
- Served as Press and Creative Writing Executive for the NUS Students' Union, crafting LinkedIn captions for events and initiatives, while developing article topics and weekly fun facts.

Volunteering Experience

- Volunteering with NUS TeachSG, working with Unity Secondary School to mentor secondary school students and support their learning and development.
- Acted as Vice-Project Director for the Elderly Service Project (Computing Club), delegating tasks to various subdepartments, organising NUH volunteering sessions, while managing proposal writing and user guides for elderly.
- Volunteered as a Python/Scratch Instructor for CITC and a Digital Literacy Instructor for SHG, teaching programming basics to students and guiding seniors in app usage, honing communication and instructional skills.

Achievements and Skills

- Achieved a top 10 placement in a national creative writing competition, leading to the publication of my debut book, now stocked in every National Library Board branch across Singapore.
- Landscape photographer, certified guitarist and passionate Tour de France fan.
- Native proficiency in English and Hindi.